Reproducing and Extending Bias Analysis in Clinical Contextual Word Embeddings: A Final Report

Ritul Soni
rsoni27@illinois.edu
University of Illinois Urbana-Champaign

Abstract

This project undertakes a reproduction study of "Hurtful Words: Quantifying Biases in Clinical Contextual Word Embeddings" by [Zhang et al. 2020], focusing on gender bias in clinical BERT models. Our objective was to reproduce key methodologies using the authors' publicly available `Baseline_Clinical_BERT` model and the full MIMIC-III dataset, with practical adaptations for a Google Colab environment. We implemented a data preprocessing pipeline processing `NOTEEVENTS.csv` (resulting in approximately 1.85 million sampled note instances after initial filtering) to incorporate actual clinical text. Bias was first assessed using prior-adjusted log-probability scores with custom templates for "diabetes" and "pneumonia", which revealed differential likelihoods for "he" versus "she" based on context. Subsequently, a downstream in-hospital mortality prediction task was developed using a 3000-note random sample from our processed MIMIC-III text subset. Embeddings extracted via ClinicalBERT were used to train a Logistic Regression classifier, achieving an AUROC of 0.6674. Fairness analysis on this task for the 'gender' attribute indicated observable disparities, such as a recall gap of -0.0571 (Male vs. Female) and a near-zero precision gap of -0.0025. Although full numerical replication of the extensive experiments of the original article was limited by dataset scale for the downstream task and template specificity, our findings support the fundamental conclusion that clinical language models can exhibit quantifiable biases. This work underscores the methodological feasibility of such audits and highlights the critical impact of data representativeness and scale on observed bias and model performance.

Project Video:	https://tinyurl.com/FinalProjectVid
Alt Video Link:	https://mediaspace.illinois.edu/media/t/1_ikioxjcw
GitHub Repository:	https://github.com/RitulSoni/CS598-Hurtful-Words-Final-Project

Introduction

The advancement of large language models (LLMs) and contextual word embeddings has significantly impacted various fields, including clinical natural language processing (NLP). These models offer powerful representations of text, enabling improvements in tasks such as diagnosis prediction, patient stratification, and information extraction from clinical notes. The paper "Hurtful Words: Quantifying Biases in Clinical Contextual Word Embeddings" by [Zhang et al. 2020]. investigates a critical concern in this domain: the extent to which these clinical embeddings may encode and perpetuate societal biases present in the training data. Their work demonstrates that BERT models pre-trained or fine-tuned on clinical notes from the MIMIC-III dataset can exhibit statistically significant disparities in performance and representation for marginalized populations concerning attributes like gender, ethnicity, and insurance status. The primary contributions of [Zhang et al. 2020] include quantifying these biases using both a template-based fill-in-the-blank method (log probability bias scores) and by evaluating performance gaps across numerous downstream clinical prediction tasks. They also explore the efficacy of adversarial debiasing techniques. This research highlights the crucial need for auditing and mitigating bias in AI systems deployed in high-stakes healthcare settings to prevent the exacerbation of health disparities.

Scope of Reproducibility

This project aimed to reproduce key methodological aspects of [Zhang et al. 2020]. Specifically, we focused on:

- **Dataset Processing:** We utilized the full MIMIC-III clinical dataset (v1.4) CSVs. A simplified data preprocessing pipeline was implemented in Python (pandas, numpy) within a Google Colab environment. This involved loading `PATIENTS.csv`, `ADMISSIONS.csv`, and `DIAGNOSES_ICD.csv` in full. Critically, `NOTEEVENTS.csv` was processed by taking an approximate 10% random sample (chunk-wise processing, resulting in 1,851,344 note rows initially filtered to obtain a manageable subset of clinical text for analysis. This allowed work with actual note content.
- **The Model:** We successfully loaded and utilized the `Baseline_Clinical_BERT` model provided by the original authors via their GitHub repository, after extracting it from its `.tar` archive. We did not reproduce the pre-training of this model.
- **Bias Quantification (Log Probability Scores):** The methodology for calculating prior-adjusted log probabil-

ity bias scores (Algorithm 1 from the paper) was reproduced. This was applied using three custom templates each for two clinical attributes (”diabetes” and ”pneumonia”) with ”he” and ”she” as target gendered pronouns.
- **Downstream Task Evaluation:** One simplified downstream binary classification task (in-hospital mortality prediction, using `hospital_expire_flag`) was implemented. For this, a random sample of 3000 notes was taken from our processed MIMIC-III text subset. Embeddings were extracted from `Baseline_Clinical_BERT` for these notes, and a Logistic Regression classifier was trained. This classifier's overall performance and fairness metrics (Demographic Parity, Equality of Opportunity for positive and negative classes, False Positive Rate Gap, and as an extension, Precision Gap) for the 'gender' attribute were calculated. This is a scaled-down version of the 57 downstream tasks in the original paper.
- **Baselines/Adversarial Debiasing:** These aspects of the original paper were outside the scope of this project due to time and computational constraints.

While not a full replication of all experiments, this project successfully reproduced core mechanisms for bias quantification presented in [Zhang et al. 2020] using their model on a substantial, text-inclusive subset of MIMIC-III.

Methodology

Environment

This project was developed in a Google Colab environment.
- **Python Version:** 3.11.12
- **Key Dependencies:**
  - `pandas` (2.2.2)
  - `numpy` (2.0.2)
  - `torch` (2.6.0+cu124)
  - `transformers` (4.51.3)
  - `scikit-learn` (1.6.1)
  - `matplotlib` (3.10.0), `seaborn` (0.13.2)
  - `tarfile` (standard library, Python 3.11.12)

Data

Data Source and Access The project utilized the MIMIC-III (Medical Information Mart for Intensive Care III) Clinical Database v1.4 [Johnson et al. 2016]. Access to the full dataset CSVs was obtained via PhysioNet credentials. For data download instructions, refer to https://physionet.org/content/mimiciii/1.4/.

Data Preprocessing The data preparation was performed in a Jupyter Notebook: `data_preprocessing.ipynb` (see the GitHub repository).

1. **Loading and Processing Notes:** The full `PATIENTS.csv`, `ADMISSIONS.csv`, and `DIAGNOSES_ICD.csv` files were loaded. The `NOTEEVENTS.csv` file was processed (e.g., via chunking and sampling, or loaded fully if memory permitted), resulting in 1,851,344 note rows initially filtered and retained (i.e., the shape of the `notes_df_filtered` DataFrame before merging).
   To efficiently sample without loading the entire file, O ised `pd.read_csv` with `chunksize`, and applying `chunk.sample(frac=0.1)`.
2. **Merging and Cleaning:** Tables were merged using `subject_id` and `hadm_id`. Column names were standardized. Patient age was computed from `admittime` and `dob`, accounting for MIMIC-III date shifts (1,790,894 valid ages).
   Ethnicity and language values were cleaned and mapped to `ethnicity_clean` and `language_clean`. ICD-9 codes were grouped into lists per admission.
3. **Final Dataset:** We retained only rows with essential values: `subject_id`, `gender`, `ethnicity_clean`, `age_at_admission`, `text_note`, and `hospital_expire_flag`.
   The final DataFrame had 1,790,301 rows. This processed dataset was saved as `processed_mimic_full_subset.csv`.

Table 1: Descriptive Statistics of the 3000-Note Sample Used for Downstream Task.

Characteristic	Value
Total Notes in Sample for Task	3000
Unique Patients in Sample	2628
Unique Admissions in Sample	2692
Median `age_at_admission` (years)	58.2
Gender Distribution (M/F in 3000 sample)	M: 1737 (57.9%), F: 1263 (42.1%)
Label (`hospital_expire_flag` in 3000 sample)	0 (Survived): 2589 (86.3%), 1 (Expired): 411 (13.7%)

Model

The `Baseline_Clinical_BERT` model from [Zhang et al. 2020] was used.
- **Repository:** https://github.com/MLforHealth/HurtfulWords
- **Description:** A BERT-base architecture [Devlin et al. 2019] initialized from SciBERT [Beltagy, Lo, and Cohan 2019] and further pre-trained on MIMIC-III clinical notes.
- **Usage:** MLM head for template-based bias scores; base model for `[CLS]` token embeddings (768-dim) for the downstream task using the `text_note` column.
- **Model Implementation/Usage:** The pre-trained model was obtained from the original authors' repository. Implementation involved `.tar` archive extraction, loading the model and tokenizer via the Hugging Face `transformers` library (using `AutoTokenizer`, `AutoModelForMaskedLM`, and `AutoModel`), generating embeddings from the `text_note` column for the downstream task, and implementing Algorithm 1 (from the original paper) for calculating template-based bias scores. This process included addressing date-related `OverflowErrors` encountered during initial execution with the dataset.

Training (Downstream Task)

Training refers to the Logistic Regression classifier for in-hospital mortality.
- **Classifier:** `sklearn.linear_model.LogisticRegression`.

- **Hyperparameters:** `random_state=42, max_iter=1000, class_weight='balanced', solver='lbfgs', C=1.0`.
- **Computational Requirements:** Google Colab (T4 GPU for embeddings). Embedding 3000 notes: $\sim$15–20 minutes. Generating full embeddings estimated time is 300 hours. Logistic Regression training: seconds.
- **Training Details:** Input: 768-dim embeddings from 2100 training notes. Target: `hospital_expire_flag`. Loss: Log-loss.

Evaluation (Metrics)

Metrics were aligned with [Zhang et al. 2020]:
- **Template-Based Bias:** Prior-Adjusted Log Probability Bias Score: $\log\left(\frac{p_i}{p_{\text{prior}}}\right)$.
- **Downstream Performance:** Accuracy, Precision, Recall, F1-Score, AUROC, AUPRC.
- **Downstream Fairness (Male vs. Female):** Demographic Parity Gap ($P(\hat{Y} = 1 \mid Z = M) - P(\hat{Y} = 1 \mid Z = F)$), Recall Gap ($\text{TPR}_M - \text{TPR}_F$), Specificity Gap ($\text{TNR}_M - \text{TNR}_F$), FPR Gap ($\text{FPR}_M - \text{FPR}_F$).

Results

Template-Based Bias Scores

For "diabetes," the mean score for "she" (6.5530) was higher than for "he" (5.6134). For "pneumonia," "he" (3.5724) was slightly higher than for "she" (3.3916) (Table 2, Figure 1). These results indicate context-dependent gendered associations, differing from [Zhang et al. 2020]'s Table 3 values (e.g., Diabetes M:0.205, F:-0.865), likely due to our small, custom template set (3 per attribute). The core observation of differential scores holds.

Table 2: Mean Prior-Adjusted Log Probability Bias Scores.

Attribute	Mean Score "he"	Mean Score "she"
Diabetes	5.6134	6.5530
Pneumonia	3.5724	3.3916

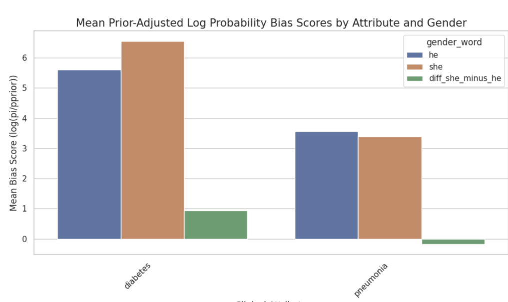

Figure 1: Mean Prior-Adjusted Log Probability Bias Scores by Attribute and Gender.

Downstream Task: In-Hospital Mortality Prediction

The Logistic Regression on 3000 sampled notes achieved an AUROC of 0.6674 and AUPRC of 0.2300 (Table 3). Recall for the positive (Expired) class was 0.41.

Table 3: Overall Performance: In-Hospital Mortality (3000 Sampled Notes).

Class	Precision	Recall	F1-Score	Support
0 (Survived)	0.89	0.77	0.83	777
1 (Expired)	0.22	0.41	0.29	123
Accuracy			0.72	900
Macro Avg	0.56	0.59	0.56	900
Weighted Avg	0.80	0.72	0.75	900
AUROC: 0.6674			AUPRC: 0.2300	

Fairness metrics for gender are in Table 4 and Figure 2. Gaps are in Table 5 and Figure 3.

Table 4: Fairness Metrics by Gender (Downstream Task).

Gender	N	Pos.Pred.Rate	Recall(TPR)	Specificity(TNR)	FPR	Precision(PPV)
F	366	0.2705	0.4490	0.7571	0.2429	0.2222
M	534	0.2472	0.3919	0.7761	0.2239	0.2197

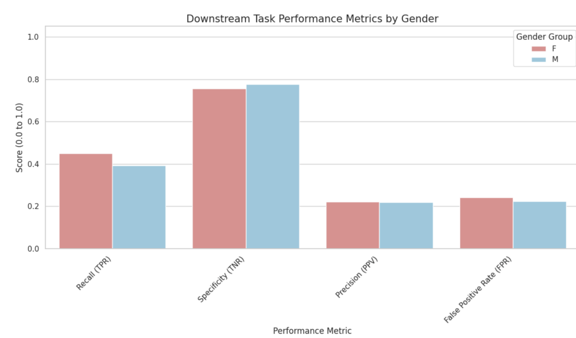

Figure 2: Downstream Task Performance Metrics by Gender.

Table 5: Fairness Gaps (M vs. F) (Downstream Task).

Gap Type	Value (M-F)
Demographic Parity Gap	-0.0233
Recall (TPR) Gap	-0.0571
Specificity (TNR) Gap	0.0190
False Positive Rate (FPR) Gap	-0.0190
Predictive Equality (Precision) Gap	-0.0025

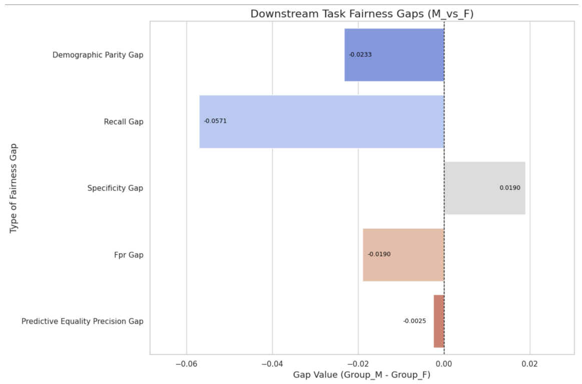

Figure 3: Downstream Task Fairness Gaps (Male - Female).

The Recall Gap of -0.0571 (M vs. F) indicates lower recall for males. The original paper [Zhang et al. 2020] found males often had higher recall in tasks with significant gender gaps; our single task differs, highlighting task/sample specificity.

Additional Extension: Predictive Parity

To further explore fairness dimensions beyond those emphasized in the original work's main downstream analysis (Recall and Specificity Gaps), we extended the evaluation by calculating the Predictive Parity (Precision Gap). This metric assesses whether the precision (Positive Predictive Value, PPV) of positive predictions differs across groups, providing insight into the reliability of a positive outcome depending on the demographic group.

The results are shown in Table 5. We found a Precision Gap (M-F) of -0.0025. This very small gap suggests that, for this specific task and data sample, the reliability of a positive mortality prediction (i.e., the likelihood that a patient predicted to expire actually did) was similar across genders, despite the previously noted Recall Gap.

Additionally, considering Equalized Odds—a stricter fairness criterion requiring near-zero gaps in both True Positive Rate (Recall) and False Positive Rate—our calculated components (TPR gap -0.0571, FPR gap -0.0190, see Table 5) demonstrate that this condition was not satisfied by the model on this task.

Discussion

Our reproduction demonstrates that the `Baseline_Clinical_BERT` exhibits context-dependent gender associations (template analysis) and that fairness gaps can be quantified in downstream tasks (mortality prediction: recall gap M-F of -0.0571, precision gap -0.0025). While specific numerical results differ from [Zhang et al. 2020] due to scale, template choices, and downstream model simplicity, our findings support their central thesis: biases are present and measurable. The modest downstream performance (AUROC 0.6674) using actual (sampled) text highlights challenges with limited data for complex tasks, yet fairness disparities were still observable.

The original paper is **partially reproducible** in its core methodologies. Full numerical reproduction is challenging due to their extensive data processing pipeline (requiring specific DB setup and high compute, which we adapted by CSV sampling), unreleased specific templates, and the scale of 57 downstream tasks. **Easy aspects:** Loading the pre-trained model (post-extraction), basic downstream task setup. **Difficult aspects:** Full MIMIC-III CSV preprocessing in Colab (date overflows, memory management for sampling `NOTEEVENTS.csv`), accurately implementing Algorithm 1 for bias scores, interpreting results from limited data. **Recommendations for Reproducibility:** Authors could release specific templates used for key bias score results, cohort definitions for a few representative downstream tasks, or more detailed, containerized/simplified processing scripts for a MIMIC sample.

Limitations and Setbacks

Several practical challenges and limitations were encountered during this reproduction study, influencing the scope and specific implementation choices:

- **MIMIC-III Data Access:** A significant initial hurdle was obtaining access to the MIMIC-III dataset. This requires completing credentialing requirements (CITI training) and submitting an application via PhysioNet, which involves administrative overhead and waiting periods, impacting the project timeline from the start.
- **Data Scale and Processing:** Handling the full MIMIC-III dataset, particularly the large `NOTEEVENTS.csv` file roughly 2 million notes, 5GB+, posed challenges within the Google Colab environment. Memory limitations and long processing times necessitated sampling the notes data (initially retaining 1.85M note rows) rather than processing the entire corpus as likely done in the original study. Furthermore, debugging date conversions and handling potential `OverflowErrors` during age calculation required careful data cleaning and adaptation. This adaptation, while necessary for feasibility, means our model's exposure to clinical text differed from the original setup.
- **Downstream Task Scope:** The original paper evaluated fairness across 57 distinct downstream tasks. Reproducing this breadth was infeasible due to the significant effort involved in defining cohorts, processing data specifically for each task, and training/evaluating 57 separate models. We focused on reproducing the methodology on a single, representative task (in-hospital mortality prediction) using a further reduced sample (3000 notes) to keep computational requirements manageable.
- **Downstream Classifier Choice (Logistic Regression vs. NN):** The original paper utilized a tuned feed-forward neural network built on concatenated embeddings from the last four BERT layers for downstream tasks. Replicating this, including hyperparameter tuning, would have substantially increased implementation complexity and computational requirements (training time, potential need for more GPU resources than readily available in free Colab tiers). Given the time constraints and the project's focus on demonstrating the bias evaluation *pipeline*, we opted for a simpler `sklearn.linear_model.LogisticRegression`

classifier trained on embeddings from only the last BERT layer. While this is a simplification and differs from the paper's approach, Logistic Regression is a standard baseline classifier, less prone to overfitting on the smaller 3000-note sample, and allowed for the successful measurement and analysis of fairness gaps derived from the ClinicalBERT embeddings within the project's scope. It serves as a suitable replacement for demonstrating the core fairness evaluation methodology.

- **Template Specificity:** The exact templates used for the log probability bias score calculation in the original paper were not available. We created custom templates (3 per attribute) for "diabetes" and "pneumonia". While this allowed us to reproduce the *method*, the limited number and difference in specific phrasing inevitably lead to numerically different bias scores compared to the original paper, limiting direct comparison of those specific values.

These limitations underscore the challenges often faced in reproducing complex machine learning studies, particularly those involving large, restricted datasets and significant computational resources. Our adaptations aimed to preserve the core methodological spirit while ensuring project feasibility.

Author Contributions

This project was completed by Ritul K. Soni.

References

Beltagy, I.; Lo, K.; and Cohan, A. 2019. SciBERT: A Pretrained Language Model for Scientific Text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 3615–3620.

Devlin, J.; Chang, M.-W.; Lee, K.; and Toutanova, K. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 4171–4186.

Johnson, A. E.; Pollard, T. J.; Shen, L.; Lehman, L.-W. H.; Feng, M.; Ghassemi, M.; Moody, B.; Szolovits, P.; Celi, L. A.; and Mark, R. G. 2016. MIMIC-III, a freely accessible critical care database. *Scientific data*, 3(1): 1–9.

Zhang, H.; Lu, A. X.; Abdalla, M.; McDermott, M.; and Ghassemi, M. 2020. Hurtful words: Quantifying biases in clinical contextual word embeddings. In *Proceedings of the ACM Conference on Health, Inference, and Learning*, 110–120.